# CHAMINDU NIPUN

AI / ML ENGINEER | BSc (Hons) COMPUTER ENGINEERING

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## Summary

AI/ML Engineer experienced in Machine Learning, Data Science, and Generative AI, specializing in LLM-powered systems, RAG pipelines, and data-driven platforms. Skilled in large-scale data processing, end-to-end AI solutions, MLOps pipelines, and cloud deployment. Passionate about building scalable AI systems that drive automation and intelligent decision-making.

## Experience

### Boffo System Labs (Pvt) Ltd — 07/2024 - 01/2025

Software Engineering Intern

- Developed and deployed scalable web applications in production, improving performance and reliability for end users.
- Built responsive front-end with React.js and high-performance back-end services using Node.js and ASP.NET Core.
- Designed and optimized MySQL, PostgreSQL and MongoDB schemas, ensuring fast queries and efficient data management.
- Collaborated in Agile teams with Azure DevOps, GitHub and ClickUp, driving CI/CD and iterative software delivery.
- Deployed and tested cloud services, ensuring stability, high performance, and smooth release cycles.

## Education

### B.Sc. Eng. (Hons) in Computer Engineering — 01/2021 - 12/2025

Faculty of Engineering | University of Ruhuna

## TECHNICAL SKILLS

- Agentic AI & Generative AI: LangChain, Langgraph, CrewAI, RAG pipelines, LLM Deployment, n8n
- AI & ML: Regression, Classification, Clustering, Feature Engineering, Model Evaluation, PyTorch, TensorFlow, scikit-learn
- Computer Vision & Natural Language Processing: OpenCV, HuggingFace Transformers, Text & Image Processing
- Data Science : Data Preprocessing, Exploratory Data Analysis (EDA), Statistical Analysis, Insights Extraction
- Full-Stack Development & Databases: React.js, Node.js, FastAPI, Spring Boot, MongoDB, MySQL, PostgreSQL, ChromaDB
- Cloud & DevOps: AWS, Docker, Git/GitHub, Azure DevOps
- Programming Languages: Python, JavaScript, Java

## Key Achievements

**Champions - Mobitel Code Blast Ruhuna Hackathon 2025**
Won First Place for the Agentic AI-Powered Workflow Automation System at the Southern AI Expo & Conference 2025

**Finalist – IEEE Sri Lanka Section FYP Arena 2025**
Selected among top national finalists for the Undergraduate Project Competition in IT & Computer Science.

**Vice Chancellor's Awards & Academic Sessions 2026 – University of Ruhuna**
Selected in Vice Chancellor's Awards 2026, highlighting academic excellence and research innovation.

## PROJECTS

### Agentic AI-Powered Workflow Automation System — 02/2025 - 11/2025

Final Year Project | Faculty of Engineering | University of Ruhuna

- Developed a multi-agent AI system (Guidance, Hall Booking, Time Table Planner Agents) automating academic workflows, reducing manual processing time by 70%.
- Implemented Guidance Agent (faculty and governance-level Agents) using Agentic RAG-based AI agents with LLMs, embeddings models and ChromaDB for contextual multi-turn document retrieval, improving accuracy and retrieval speed.
- Built Hall Booking Agent and Time Tables Planner Agent with K-Means clustering and Google OR-Tools, optimizing scheduling for 100+ events.
- Integrated AssemblyAI and Tavily API to enable voice interaction and real-time web augmentation, enhancing accessibility and reasoning capabilities.
- Developed modular FastAPI and Node.js microservices, containerized with Docker, and automated deployments via GitHub Actions CI/CD, ensuring scalable and reliable production-ready services.
- **Tech Stack: Python, FastAPI, Node.js, React, TypeScript, LangGraph, LangSmith, LLMs, EasyOCR, AssemblyAI, Tavily API, Docker, MySQL, MongoDB, ChromaDB, Google OR-Tools.**

### PrometheusAI Data Engine

10/2025 - 11/2025

Automated Data Engineering & Analytics Platform | [Live Demo]

- Developed an Agentic AI-powered data engine automating data cleaning, visualization, and natural-language analytics.
- Built a conversational AI assistant with LLMs (Gemma-3/OpenAI via OpenRouter) for automated data querying and visualization using Matplotlib and Seaborn.
- Designed an end-to-end data pipeline with schema inference, missing-value imputation, and IQR/Z-score outlier handling for high-integrity datasets.
- Optimized memory performance by 30–70% through data type downcasting and resource-efficient CSV processing.
- Deployed modular FastAPI backend and Streamlit frontend, enabling interactive and scalable analytics workflows.
- **Tech Stack: Python, FastAPI, Streamlit, Pandas, NumPy, SciPy, Matplotlib, Seaborn, LLMs.**

### Multiple Disease Prediction System

09/2025 - 10/2025

Machine Learning & MLOps Project | [GitHub]

- Developed a multi-disease prediction platform (Diabetes, Heart Disease, Parkinson's) with trained scikit-learn models deployed across multiple environments.
- Built an API-driven FastAPI backend with SwaggerUI docs and schema validation, enabling robust and reliable inference.
- Implemented 7 end-to-end deployment pipelines (FastAPI, Docker, Heroku, Azure, HuggingFace, Ngrok, Google Colab) to demonstrate MLOps scalability.
- Deployed Streamlit dashboards for interactive user experience and containerized services with Docker for portability and reproducibility.
- **Tech Stack: Python, FastAPI, Streamlit, Docker, SwaggerUI, Azure, HuggingFace, Heroku, Google Colab, Ngrok, scikit-learn.**

### Autonomous Tech Blog Generation

04/2025 - 05/2025

Multi AI Agent System Project

- Built an autonomous multi-agent system for researching and generating tech blog articles.
- Designed sequential agents (Senior Researcher & Writer) using CrewAI and Gemini/GPT LLMs for information gathering and content generation.
- Implemented real-time web search and asynchronous execution, producing readable, markdown-formatted articles.
- **Tech Stack: Python, CrewAI, LLMs, LangChain, Serper.dev**

### Traffic Sign Recognition System

06/2024 - 07/2024

Deep Learning & Computer Vision Project

- Developed a real-time traffic sign recognition system using CNNs for autonomous driving assistance.
- Integrated OpenCV and TensorFlow to process live webcam feeds, classifying multiple signs with high accuracy.
- Achieved robust detection under varied lighting and camera angles, providing actionable driver feedback.
- Visualized detected signs directly on video streams for intuitive, real-time output.
- **Tech Stack: Python, TensorFlow, OpenCV, CNN, Computer Vision.**

## CERTIFICATIONS

- Champions - Mobitel Code Blast Ruhuna Hackathon 2025
- TensorFlow for AI, Machine Learning and Deep Learning - DeepLearning.AI
- Math for Machine Learning - Simplilearn
- Introduction to Machine Learning with Python - IBM
- Effective Problem-Solving and Decision-Making - University of California

## REFERENCES


### Dr. Thilina Weerasinghe

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